LEGAL RESEARCH AND EVIDENCE ASSISTANT – JURIAID

Project ID: 25-26J-274

PROJECT PROPOSAL REPORT

Sammani W.M.W.L.

BSc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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29/08/2025

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29/08/2025

Abstract

This project addresses a significant challenge faced by legal practitioners: the time-consuming and manual process of creating legally relevant questions (for acceptance and resolutions) during case management. Currently, there is no dedicated platform to automate this crucial step, leading to inefficiencies and increased workload for lawyers.

Our proposed solution is an AI-powered legal assistant system designed to generate these questions automatically. The system will operate by taking an initial case input, such as a complaint and the victim's reply, and then leveraging legal-specific knowledge bases (past cases, statutes, and laws) to formulate precise and contextually appropriate follow-up questions.

The core design involves fine-tuned legal language models (e.g., Legal-BERT variants), Retrieval-Augmented Generation (RAG), and context-aware fact elicitation. This will enable the system to identify missing legal facts, generate dynamic clarifying questions, and produce legally formatted outputs grounded in authoritative sources.

By automating this process, the research aims to significantly enhance the efficiency and accuracy of legal case management, reduce the manual effort involved, and provide lawyers with a powerful tool for delivering more personalized and effective legal services. The system will prioritize reliability, security, usability, and scalability to ensure practical applicability within the legal domain.

Keywords: Legal AI, Question Generation, Retrieval-Augmented Generation, Legal- BERT.

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1. INTRODUCTION

In modern legal practice, managing case files efficiently is paramount. The initial stages of a legal case involve a client providing a complaint to their lawyer, who opens a case file. A copy is sent to the defendant, whose lawyer provides a reply. A critical subsequent step requires lawyers to manually create a series of "acceptance" and "resolution" questions to gather further details, clarify ambiguities, and strategize.

However, this manual process is highly time-consuming and lacks a dedicated, automated platform [1]. Lawyers spend considerable effort drafting these questions, effort that could be allocated to complex legal analysis. This manual effort also introduces variability in the completeness and precision of the questions, potentially prolonging the legal process.

This research project proposes to develop JuriAid, an AI-powered system to address this critical need by automating the generation of these legal questions. The system will analyze the initial complaint and reply, and by leveraging past cases and relevant laws, it will intelligently formulate comprehensive and context- aware questions. Our aim is to significantly uplift the efficiency and effectiveness of legal case management.

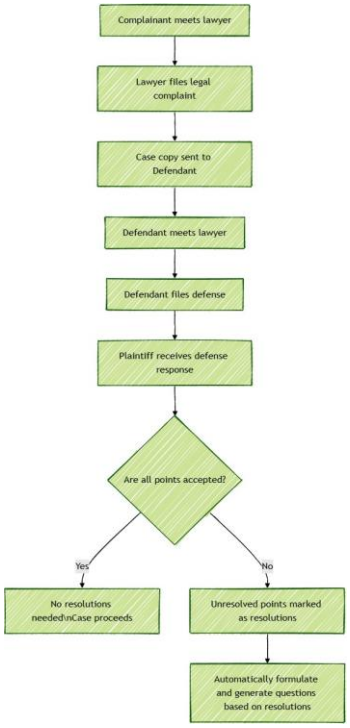

Figure 1.1 flowchart of the scenario

2. LITERATURE REVIEW

The legal domain is increasingly adopting Information and Communication Technologies (ICTs) to enhance efficiency and access to justice [1]. Traditional legal workflows, heavily reliant on manual research and document drafting, are labor-intensive and time-consuming, which highlights the necessity for AI- assisted solutions. Domain-specific models such as Legal-BERT [2] have demonstrated superior performance over generic BERT models by capturing the nuances of legal vocabulary and structure. While Legal- BERT excels in comprehension and classification tasks, it is limited in its ability to generate dynamic, context-aware questions for initial case management, illustrating a gap between text understanding and interactive fact elicitation.

To address issues of factual accuracy and grounding, Retrieval-Augmented Generation (RAG) [3] [4] architectures have been employed, effectively combining dense retrieval with generative models to produce outputs grounded in verified knowledge bases. The implementation of RAG in systems like SCaLe-QA [3] in Sri Lanka demonstrates that domain-tuned embeddings and semantic chunking can enhance retrieval performance in a localized legal context. Although these systems strengthen factual reliability, they focus on question answering rather than the iterative, client-interactive process necessary for case intake, highlighting a functional limitation when compared to the needs of dynamic question generation.

Complementing comprehension and grounding, research on interactive fact elicitation and automated drafting provides insight into simulating aspects of lawyer-client interactions. For example, Lin and Cheng (2024) [5] illustrate that fine-tuned language models can produce structured judicial documents while generating clarifying questions based on incomplete contexts. This capability suggests that AI can partially replicate human follow-up questioning, bridging the gap between static document generation and dynamic fact-gathering. However, existing models prioritize formal document output rather than the iterative elicitation process, revealing an unaddressed need for a system that integrates comprehension, factual grounding, and interactive questioning in a single workflow.

Examining these approaches comparatively, it becomes evident that each address only a part of the broader challenge. Legal-BERT [2] ensures domain-specific understanding, RAG guarantees factual grounding, and fact-elicitation models enable question generation [5].

Individually, they provide robust functionality; collectively, however, no existing system integrates these capabilities into a unified tool for automating initial case intake within the Sri Lankan legal context. Systems like SCaLe-QA [3] excel at question answering but lack generation capabilities, whereas drafting-focused approaches [5] automate document creation but do not support interactive questioning. This cross-analysis underscores the novelty and relevance of JuriAid, which aims to synthesize these proven technologies into an integrated solution that addresses a clearly defined workflow gap, enhancing efficiency, accuracy, and comprehensiveness in early-stage legal procedures.

Table 2.1 table of analysis

Aspect	Traditional Legal Processes	Existing AI Approaches	Strengths	Limitations / Gaps	Implications for JuriAid
Research & Document Drafting	Manual review of case law, statutes, and prior judgments; labor-intensive and time-consuming.	Legal-BERT [3]enables domain-specific understanding through pre-training on legal corpora.	Captures nuanced legal vocabulary; superior performance on legal downstream tasks.	Focused primarily on comprehension and classification; does not automate question generation.	Provides the semantic foundation for JuriAid to interpret legal context accurately.
Information Retrieval & Knowledge Grounding	Lawyers manually consult multiple sources to verify facts.	Retrieval-Augmented Generation (RAG) [3] [4]; systems like SCaLe-QA (Sri Lanka) leverage semantic embeddings and chunking of case law.	Ensures generated content is factually grounded; mitigates hallucinations.	Primarily applied to question answering, not interactive or iterative question generation.	RAG forms the backbone of JuriAid's knowledge grounding, enabling accurate, verifiable questions.
Automated Question & Fact Elicitation	Lawyers ask follow-up questions to fill gaps in case details.	Interactive fact elicitation models [5] and format-aware drafting (Lin & Cheng, 2024) enable generation of clarifying questions or structured documents.	Models can simulate lawyer-client interactions and draft legal outputs automatically.	Focused on structured document drafting rather than dynamic, iterative interview-style question generation.	JuriAid integrates this approach to produce dynamic, context-aware initial case management questions.
Integration of Technologies	Highly fragmented; humans coordinate comprehension,	Individual models (Legal-BERT, RAG,	Each component is highly	No single system integrates understanding, grounding, and	JuriAid synthesizes these technologies into a

	retrieval, and questioning.	fact-elicitation LLMs) exist.	effective in isolation.	question generation in a Sri Lankan context.	cohesive system tailored for automated case question generation, addressing a specific workflow gap.
Research Context	Global legal AI research often focuses on English-speaking or high-resource jurisdictions.	Some local adaptation exists (SCaLe-QA in Sri Lanka) but limited to QA tasks.	Localized models demonstrate the feasibility of adapting AI to specific legal systems.	Lack of integrated, domain-specific solutions for initial case intake and dynamic question generation.	Highlights the novelty and practical relevance of JuriAid in addressing an unmet need in Sri Lanka's legal system.

3. RESEARCH GAPS

A research gap is an area within a field of study that has not been thoroughly investigated or addressed in previous research, representing a missing element in the existing knowledge [1].

The major research gap identified in current legal practice is the absence of a dedicated and intelligent automated platform for generating legally relevant questions (acceptance and resolutions) based on case file details and legal precedents [1]. Lawyers currently rely on a manual and time-consuming process to formulate these questions, a problem explicitly stated by legal professionals.

Existing legal tech solutions may offer general legal research tools or document automation for standard templates, but they often lack the sophisticated, context-aware intelligence required to dynamically generate specific, targeted questions for individual case files by integrating historical case data and statutes in an integrated manner [3] [1]. This leaves a significant void in automating a critical, repetitive, and expertise- driven task in the legal workflow. The current reliance on manual question creation means there's no systematic way to ensure comprehensive coverage, consistency across cases, or rapid adaptation to new legal nuances without substantial human effort. This gap prevents lawyers from fully leveraging technology to enhance efficiency in the initial and ongoing stages of case assessment [1].

4. RESEARCH PROBLEMS

Lawyers currently spend significant time manually generating legally relevant questions during case preparation, a process that is both inefficient and prone to human error. The absence of an automated mechanism increases workload and delays case management. Hence, there is a critical need for an AI-based system capable of automatically generating contextually accurate and legally valid questions to improve efficiency, consistency, and overall quality in legal practice.

Based on the identified research gap, the core problems this project seeks to resolve are:

• **How to accurately and efficiently generate legally relevant questions (acceptance and resolutions) for case files?**

The current manual process is time-consuming and prone to variations in completeness [1]. The challenge lies in developing an AI system that can ingest diverse legal case details (e.g., complaint, victim's reply) and then intelligently formulate precise, structured questions that cover all necessary legal facts for clarity and resolution

• **How to ensure the generated questions are contextually appropriate, legally sound, and comprehensive, mimicking a lawyer's expertise?**

Generic AI models often struggle with the nuances of legal language and context [2]. The problem involves designing the system to integrate domain-specific knowledge, adhere to legal standards (e.g., IRAC structure implicitly), and anticipate missing information to generate follow-up questions effectively.

• **How to effectively leverage vast amounts of past case data and legal statutes to inform the question generation process?**

Lawyers currently rely on their experience and manual research of precedents. The challenge is to create a robust retrieval and reasoning mechanism that can quickly access and apply relevant legal authority and historical case patterns to the automated question generation, ensuring questions are grounded in established legal principles and precedents [3] [6]

5. OBJECTIVES

Main Objectives

The main objective of this research is to **develop an AI-powered legal assistant system** that automates and enhances the generation of context-aware, legally sound questions (acceptance and resolutions) for effective legal case management, thereby **significantly improving the efficiency and accuracy** for legal practitioners in Sri Lanka and potentially beyond. This innovation intends to better support lawyers in streamlining their workflow, reducing manual effort, and ensuring comprehensive information gathering throughout the case process.

Sub Objectives

To achieve the main objective, the following specific sub-objectives must be accomplished:

• **To design and implement a fact-elicitation module** capable of analyzing initial case inputs (complaints, replies) to identify and automatically elicit missing key legal facts (e.g., dates, parties, clauses) required for comprehensive question generation. [5]

• **To develop a grounded legal question generation engine** that leverages retrieval-augmented large language models (LLMs) to produce precise questions backed by relevant statutes, case law, and precedents, thereby reducing inaccuracies and improving reliability. [2] [4] [3]

• **To create an intelligent document retrieval system** capable of fetching relevant statutes, precedent snippets, and legal documents via semantic retrieval systems (e.g., Legal-BERT embeddings) and lexical methods (e.g., BM25) to inform the question generation process. [3] [7]

• **To provide a user-friendly interface for lawyers** to input case details, review generated questions, and interact with the system for clarifications and refinements, ensuring seamless integration into existing legal workflows.

6. METHODOLOGY

The development of this AI-powered legal question generation system will follow a systematic methodology, encompassing data collection, model development, system integration, and rigorous testing. The overall system aims to provide a comprehensive solution for legal question automation.

Overall System

The whole system, as described by your context, can be conceptualized as an AI-powered legal assistant with a core component for question generation. Referring to the provided system diagram, the overall architecture of such a legal AI system would include: [3] [7]

• User Input: Lawyers provide initial case details, complaints, and replies.

• Agent Framework Orchestrator: Manages task distribution and overall workflow.

• Court Decision Retrieval & Law Retrieval Engine: These components are responsible for fetching relevant statutes, precedents, and case law from a comprehensive legal knowledge graph (e.g., Neo4j) and embedding stores (using FATSS/SBERT and embeddings).

• Questionnaire Generator (Core Component): This module, powered by fine-tuned Legal-BERT with LoRA adaptation, dynamically generates context-aware questions and assesses legal positions.

• Legal Position Evaluation & Report Generator: Assesses the legal position based on input and generated questions, leading to output synthesis (e.g., draft questions, summaries).

• Long-Term Memory: Stores query history and aids in continuous learning.

• Final Output: Comprehensive legal analysis, guidance, and actionable recommendations (in your case, the generated legal questions).

Individual Component Diagram (Legal Question & Q&A Component)

The core "Legal Questionnaire & Q&A" component, which focuses on generating questions, operates through a pipeline: [3] [5]

1. Input: User (lawyer) submits initial case details (e.g., a complaint, reply).

2. Retrieval: The system uses Legal-BERT embeddings to query case-law and statute indexes, retrieving relevant chunks of legal information. This is done via a domain-tuned retrieval

system.

3. Analysis: A fine-tuned LLM (e.g., Legal-BERT-based) analyzes the user input along with the retrieved legal information. If crucial facts are missing, it generates targeted, clarifying questions for fact elicitation.

4. Follow-Up: The user provides answers to the generated questions, and the LLM may iterate by asking further clarifying questions until sufficient context is gathered.

5. Generation: Once enough context is available, the LLM generates the final output, which in your system would be the legally formatted "acceptance and resolution" questions, citing relevant authorities.

6. Feedback: The system may prompt for review or flag areas needing validation from the lawyer.

Key Tasks Involved:

• Fact Elicitation: Assessing user input for missing key legal facts and auto-generating focused follow-up questions. [5]

• Question Generation: Producing structured, legally relevant questions (e.g., "Date of incident?", "Jurisdiction?").

• Document Retrieval: Fetching statutes and precedent snippets via semantic retrieval systems (LEGAL- BERT embeddings) and lexical BM25 methods.

• Answer Generation / Drafting: Using retrieved documents as prompt context for generating grounded responses or draft documents, maintaining legal formatting (e.g., the final questions for lawyers).

• Feedback & Refinement: Iterative cycles where the LLM may ask for clarifications or regenerate text based on new inputs or system checks.

Technology Stack: The project will utilize Legal-BERT variants and other specialized LLMs fine-tuned on legal corpora, Retrieval-Augmented Generation (RAG) frameworks, potentially knowledge graphs (e.g., Neo4j) for structured legal information, and advanced NLP techniques for text processing and semantic understanding. The system will also handle multilingual inputs, specifically Sinhala and English, as relevant to the Sri Lankan legal context. The user interface for lawyers would likely be a web application (e.g., using Next.js for therapists in other similar projects).

Figure 6.1 system diagram

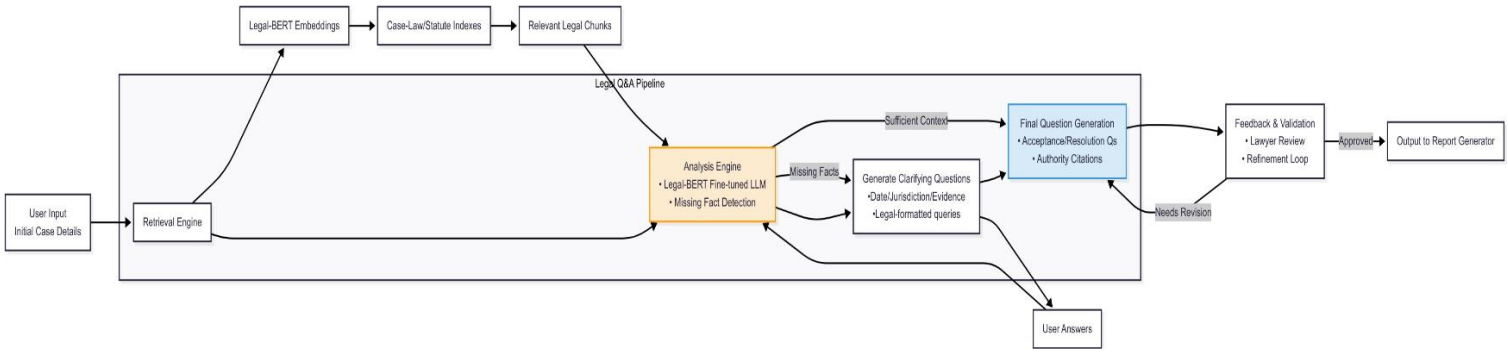

Figure 6.2 component diagram

7. PROJECT REQUIREMENTS

Functional Requirements

Functional requirements outline the precise actions, duties, or responsibilities that the system must carry out. For the legal question generation component, these include:

• Case Input Processing: The system shall accept diverse inputs related to a legal case, including initial complaints, victim replies, and other relevant case details from the lawyer.

• Fact Elicitation: The system shall analyze the provided case input to identify and highlight missing crucial legal facts (e.g., names, dates, locations, specific clauses). [5]

• Dynamic Question Generation: The system shall automatically generate a series of legally relevant and context-aware questions (acceptance and resolutions) based on the processed case details and identified missing facts.

• Legal Knowledge Retrieval: The system shall retrieve relevant legal statutes, precedents, and past case decisions from a comprehensive legal knowledge base to ground the generated questions in authoritative sources. [3] [6]

• Contextual Adaptation: The system shall adapt the generated questions based on iterative feedback from the lawyer, allowing for refinement and deeper inquiry into specific aspects of the case.

• Output Formatting: The system shall present the generated questions in a clear, structured, and legally appropriate format, suitable for direct use by lawyers. [5]

• Multilingual Support: The system should support processing inputs and generating questions in multiple languages relevant to the legal context, such as Sinhala and English.

Non-Functional Requirements

Non-functional requirements define the criteria that describe the system's behavior and quality attributes.

• Security: The system must implement robust security measures to protect sensitive client information and confidential legal documents from unauthorized access, breaches, and threats.

• Reliability: The system must be highly reliable, ensuring consistent availability for lawyers to use at any time, with minimal downtime or errors during question generation.

• Efficiency: The system must significantly reduce the time lawyers spend on manually generating questions, ensuring quick processing of inputs and rapid generation of questions. [2]

• Usability: The system's interface must be intuitive and user-friendly for legal practitioners, allowing for easy input of case details and straightforward review of generated questions.

• Scalability: The system must be designed to accommodate a growing number of users (lawyers) and an increasing volume of case data without compromising performance or response times.

• Accuracy: The generated questions must be legally accurate, relevant, and comprehensive, reflecting the nuances of legal reasoning and complying with legal standards. [2] [3]

• Explainability: Where possible, the system should provide insights into the reasoning behind generated questions or highlight the legal sources that informed its output, enhancing user trust (implicitly covered by "grounded generation" and "citation-aware" features).

8. COMMERCIALIZATION

Budget

The following amounts are estimated and may be subject to change:

Table 8.1 budget table

Description	Cost	Occurring
Server Hosting	Covered by GitHub Pack	Monthly
Cloud Database (e.g., AWS RDS for legal data storage)	Covered by GitHub Pack	Monthly
Legal Materials	LKR 10,000.00	Annual
Developer Tools/Software Subscriptions	Covered by GitHub Pack	Monthly/Annual
LLM API Usage	LKR 10,000.00	Monthly

Commercialization Methods

It is advised to launch this system as a **stand-alone entity** rather than being bound to a cooperative organization, especially given the current lack of a centralized, efficient platform in Sri Lanka that connects legal professionals with such AI tools. This approach ensures that any lawyer, regardless of their firm or affiliation, can utilize the system.

The system will be designed for **multi-device support**, with a dedicated **mobile application for lawyers** to access its features. It will likely adopt a **freemium business model**. A free tier could offer basic question generation functionalities or a limited number of questions per case. The premium tier would unlock advanced features crucial for comprehensive legal practice, such as:

- **Unlimited, sophisticated question generation** with deeper legal context and nuance.
- **Integration with broader legal knowledge bases** and real-time legal updates.
- **Advanced fact elicitation and iterative refinement capabilities.**

9. GRANTT CHART

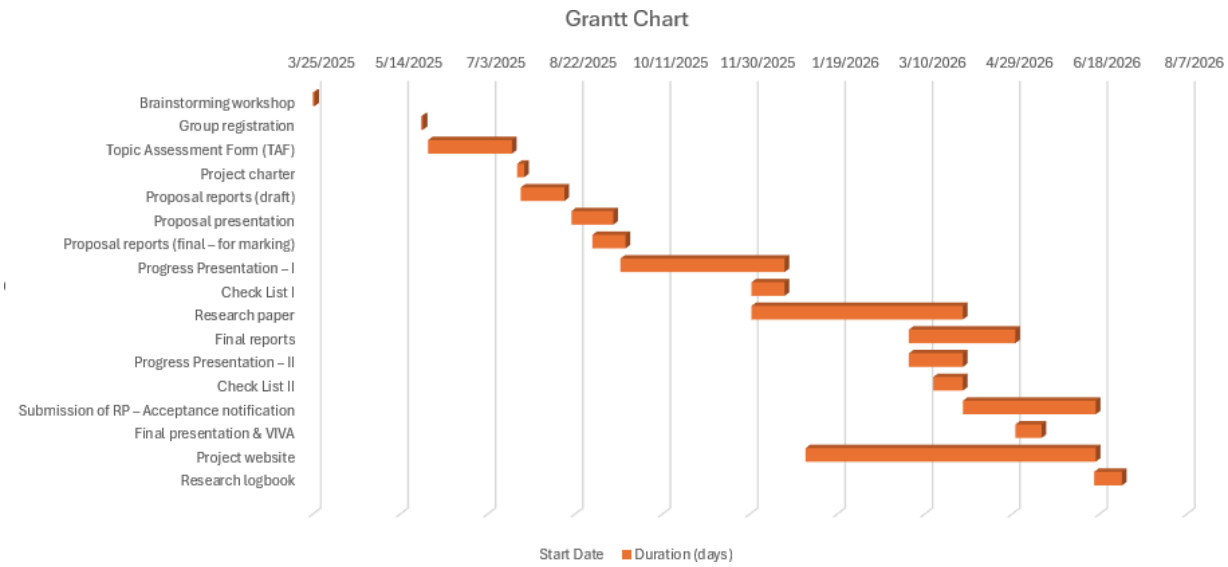

Figure 9.1 Grantt chart

10. Work Breakdown Structure

Figure 10.1 WBS chart

11.REFERENCES

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12.Appendices

IT22318848_Proposal_Report_Draft.pdf

ORIGINALITY REPORT

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Appendix A: similarity in Turnitin